A Logarithmic Phase-Walk Observable for Riemann Zeta Zeros and Riemann–von Mangoldt Backgrounds

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Abstract

We introduce a logarithmic phase-walk observable for the ordinates of the nontrivial zeros of the Riemann zeta function. The construction preserves the natural order of the zero sequence and applies an exponential phase weight depending on the index. The smooth comparison sequence is obtained by Lambert W inversion of the mean Riemann–von Mangoldt zero-counting law, giving the Riemann–von Mangoldt background (RVM). The transform separates the smooth counting trend from the arithmetic phase fluctuations of the true zeta zeros. The principal observation is that the normalized one-step complex gaps of the transformed zeta–RVM residual follow a Rayleigh law. This Rayleigh behaviour occurs at a different statistical level from the Montgomery–Odlyzko GUE law: GUE describes spacings of zero ordinates, whereas the Rayleigh law appears in the radial distribution of transformed residual increments. The paper reports this Rayleigh behaviour as a newly detected statistical feature of zeta-zero residuals under a logarithmic phase-walk transform. The same observable also distinguishes the confined RVM control from the coherent finite-window morphology of the true zeta sequence.

Keywords: Riemann zeta function; Riemann–von Mangoldt formula; Lambert W; zeta zeros; spectral statistics; quantum chaos; exponential sums; window amplitudes; GUE.

MSC 2020: 11M06; 11M26; 11Y35; 81Q50; 65C05.

1 Introduction

The Riemann zeta function and its nontrivial zeros are central objects in analytic number theory. We write a nontrivial zero as

$$\rho_k = \sigma_k + i\gamma_k, \tag{1.1}$$

where the ordinates are ordered increasingly. Under the Riemann Hypothesis one has

$$\sigma_k = \frac{1}{2} \tag{1.2}$$

for every nontrivial zero. The numerical phase-walk studied here uses tabulated critical-line ordinates and evaluates the transformed terms with amplitude parameter

$$\sigma = \frac{1}{2}. \tag{1.3}$$

We use the spectral notation

$$E_k = \gamma_k. \tag{1.4}$$

The connection between zeta zeros and spectral statistics is one of the most important bridges between number theory and mathematical physics. Montgomery's pair-correlation work connected the local statistics of zero ordinates with the two-point correlation law later identified with the Gaussian Unitary Ensemble, and Odlyzko's computations supplied extensive numerical evidence for this picture [5, 6, 8]. These classical statistics are local or correlation-based. They do not directly measure how an ordered zero sequence behaves after a nonlinear exponential phase projection.

The present paper studies such an ordered projection. Define the logarithmic phase weight

$$\tau_k = \pi \frac{\log\log k}{\log k}, \qquad k \geq 3, \tag{1.5}$$

and form

$$z_k^{\zeta} = \exp\left(\frac{\tau_k}{2} + iE_k\tau_k\right). \tag{1.6}$$

The ordered phase-walk amplitude is

$$S_N^{\zeta} = \sum_{k=3}^{N} z_k^{\zeta}. \tag{1.7}$$

The word ordered is essential: the observable is not the unordered cloud of points z_k^{ζ}, but the cumulative interference generated by the zero sequence in its natural order.

The smooth comparison sequence is obtained from the inverse Riemann–von Mangoldt counting approximation. In the computation, this inverse background is evaluated through the corresponding Lambert W inversion of the mean zero-counting law. We denote the resulting sequence by RVM and write its ordinates as

$$E_k^{(0)} = \gamma_k^{\mathrm{RVM}}. \tag{1.8}$$

Thus, throughout the paper, RVM refers to the Lambert W-based inverse Riemann–von Mangoldt background.

The aim of the article is computational and structural. We compare the zeta walk (1.7) with the same transform applied to RVM, study the residual sequence, and test local and window-level statistics. The main conclusion is that the observable provides a complementary spacing-sensitive characterization: it is consistent with the usual GUE local-spacing picture and the Riemann–von Mangoldt counting asymptotics, but it displays mesoscopic and macroscopic phase organization that those methods do not show.

2 Materials and methods

2.1 Standard background and status of the present observable

The Riemann–von Mangoldt formula, the interpretation of γ_k as zero ordinates, and the use of unfolded gaps are standard material in analytic number theory; see [2–4]. The relation between zero statistics and random matrices is literature-based material; see Montgomery and Odlyzko [5, 6], Katz–Sarnak and Rudnick–Sarnak [11, 12], and Keating–Snaith and Conrey–Snaith [13, 14]. The logarithmic phase-walk observable defined below is the original object studied in this article.

Definition 2.1 *The classical locally unfolded zeta gap is a quantity of the form*

$$s_k^{\mathrm{cl}} = \frac{\gamma_{k+1} - \gamma_k}{\langle \gamma_{k+1} - \gamma_k \rangle_{\mathrm{loc}}}. \tag{2.1}$$

It is a nearest-neighbor spacing statistic.

The observable in (1.7) is not the statistic in (2.1). It tests the ordered phase field

$$\theta_k^{\zeta} = E_k \tau_k, \tag{2.2}$$

with τ_k as in (1.5). Consequently, agreement with GUE at the level of local spacings does not force the logarithmic phase-walk observable to be locally GUE-distributed.

2.2 Position relative to known exponential sums

Standard exponential sums over zeros enter the explicit formula and related Landau–Gonek type formulae; see [2, 4]. Other phase statistics, such as fractional parts of linear forms in zero ordinates, have been studied in connection with pair correlation and prime distribution questions; see [7]. Random-matrix observables test unfolded local spectral statistics; see [5, 6, 8].

The observable introduced here differs from the standard exponential-sum, fractional-part, and random-matrix contexts described above. It is not a fixed-parameter sum of the form $\sum e^{i\gamma t}$, not an explicit-formula sum over x^{ρ}, not a fractional-part statistic of $\alpha\gamma$, and not an unfolded spacing statistic. It is an ordered cumulative sum of the phase field $\gamma_k \tau_k$, with $\tau_k = \pi \log\log k / \log k$, compared against the same transform applied to the RVM background. Within the cited literature, no comparable publication is known to the author in which a single exponential-sum transform reveals a morphology of the present complexity while sharply separating the RVM background from the zeta response. This motivates treating (1.7) as the first member of a broader family of ordered zeta observables.

Heuristic 2.2 *This is a heuristic interpretation. The observed lower nucleus, arm, upper entry, and tube are not arbitrary plotting artifacts. They indicate that the logarithmic clock imposes a susceptibility skeleton in k, and that the zeta ordinates excite this skeleton through ordered phase correlations. The visible structures therefore suggest an underlying law governing block coherence in the phase field $\gamma_k \tau_k$. Standard GUE spacing statistics are not designed to detect this law, because they discard the ordered phase-walk interference encoded in S_N^{ζ} and $B_W^{\zeta}(k)$.*

2.3 Logarithmic phase-walk observable

Definition 2.3 *For $k \geq 3$, define the logarithmic proper time*

$$\tau_k = \pi \frac{\log\log k}{\log k}. \tag{2.3}$$

The lower bound $k \geq 3$ is only a domain convention: it ensures that $\log\log k$ is real and positive. It has no independent spectral meaning.

Definition 2.4 *Let $E_k = \gamma_k$, and let σ be a real amplitude parameter. The zeta logarithmic phase-walk term is*

$$z_k^{\zeta}(\sigma) = \exp\left((\sigma + iE_k)\,\tau_k\right). \tag{2.4}$$

The finite computations in this article use

$$z_k^{\zeta} = z_k^{\zeta}\left(\frac{1}{2}\right) = e^{\tau_k/2} e^{iE_k\tau_k}. \tag{2.5}$$

Definition 2.5 *Let $E_k^{(0)} = \gamma_k^{\mathrm{RVM}}$ denote the Lambert-based RVM background ordinate. For a real amplitude parameter σ, the background term is*

$$z_k^{\mathrm{RVM}}(\sigma) = \exp\left(\left(\sigma + iE_k^{(0)}\right)\tau_k\right). \tag{2.6}$$

The finite computations in this article use

$$z_k^{\mathrm{RVM}} = z_k^{\mathrm{RVM}}\left(\frac{1}{2}\right). \tag{2.7}$$

Definition 2.6 *The residual transformed term is*

$$d_k = z_k^{\zeta} - z_k^{\mathrm{RVM}}. \tag{2.8}$$

Definition 2.7 *The cumulative amplitudes are*

$$S_N^{\zeta} = \sum_{k=3}^{N} z_k^{\zeta}, \tag{2.9}$$

$$S_N^{\mathrm{RVM}} = \sum_{k=3}^{N} z_k^{\mathrm{RVM}}, \tag{2.10}$$

$$S_N^{\mathrm{diff}} = \sum_{k=3}^{N} d_k. \tag{2.11}$$

By the definition of d_k, the finite cumulative sums satisfy the algebraic decomposition

$$S_N^{\zeta} = S_N^{\mathrm{RVM}} + S_N^{\mathrm{diff}}. \tag{2.12}$$

This identity is bookkeeping, not an explanatory theorem. The curve under study is S_N^{ζ}. RVM is a diagnostic control used to expose which features are imposed by the logarithmic clock and which features are strongly excited by the true ordinates.

2.4 Spectral interpretation of the phase walk

Heuristic 2.8 *This is a heuristic interpretation. Under the spectral identification* (1.4)*, the phase of the transformed term is*

$$\theta_k^{\zeta} = E_k \tau_k. \tag{2.13}$$

Thus (2.5) *may be read as a spectral amplitude evaluated in a non-uniform phase-walk gauge. The factor $e^{\tau_k/2}$ is a slowly varying amplitude, while the phase $E_k\tau_k$ carries the main interference structure.*

A standard spectral trace or return amplitude would use a common time variable and terms of the form $e^{iE_k t}$. The present observable uses the level-dependent time τ_k. Therefore it tests ordered phase interference rather than only the distribution of spacings.

2.5 Semiclassical background and residual phase response

Definition 2.9 *The spectral defect relative to the RVM background is*

$$\delta E_k = E_k - E_k^{(0)}. \tag{2.14}$$

The residual has the exact factorization

$$d_k = z_k^{\mathrm{RVM}} \left(e^{i\delta E_k \tau_k} - 1 \right). \tag{2.15}$$

This identity isolates the relevant fluctuation δE_k, but it is only algebraic. The analytical question is whether the ordered sequence of these phase defects has correlations beyond a chosen null model.

Heuristic 2.10 *This is a heuristic argument. If the phase defect $\delta E_k \tau_k$ is small enough to linearize the exponential, then*

$$e^{i\delta E_k \tau_k} - 1 \approx i\delta E_k \tau_k. \tag{2.16}$$

Combining (2.16) *with* (2.15) *gives*

$$d_k \approx i z_k^{\mathrm{RVM}} \delta E_k \tau_k. \tag{2.17}$$

This gives the physical interpretation of the residual as a first-order phase response of the logarithmic phase-walk observable to the spectral defect δE_k.

The spectral defect $\delta E_k = E_k - E_k^{(0)}$ is not a new source of randomness. It should be compared with the classical oscillatory term in the Riemann–von Mangoldt counting formula. If

$$N(T) = N_0(T) + S(T) + O(T^{-1}), \tag{2.18}$$

where

$$S(T) = \frac{1}{\pi} \arg \zeta \left(\frac{1}{2} + iT \right), \tag{2.19}$$

then the smooth inverse ordinate $E_k^{(0)}$ is defined heuristically by $N_0(E_k^{(0)}) = k$, whereas the true ordinate satisfies $N(\gamma_k) = k$. Linearizing (2.18) gives

$$\delta E_k = \gamma_k - E_k^{(0)} \approx -\frac{S(E_k^{(0)})}{N_0'(E_k^{(0)})}. \tag{2.20}$$

Since

$$N_0'(T) = \frac{1}{2\pi} \log \frac{T}{2\pi}, \tag{2.21}$$

this gives the heuristic scale relation

$$\delta E_k \approx (-1)\, \frac{2\pi S(E_k^{(0)})}{\log(E_k^{(0)}/2\pi)}. \tag{2.22}$$

Thus the zeta–background ordinate defect is essentially the zero-counting fluctuation $S(T)$, rewritten as an ordinate displacement.

Heuristic 2.11 *This is a heuristic mechanism. Selberg's central limit theorem implies Gaussian behaviour for the logarithm of the zeta function on the critical line; in particular, its imaginary part, and therefore the phase term underlying $S(T)$, has variance of order $\log\log T$, with normalization depending on the convention for the argument [17–19]. Combining this with* (2.22) *indicates that δE_k is a smoothed version of the Selberg-scale counting fluctuation. The logarithmic phase-walk phase defect is therefore*

$$\tau_k \delta E_k \approx (-1)\, \pi \frac{\log\log k}{\log k} \frac{2\pi S(E_k^{(0)})}{\log(E_k^{(0)}/2\pi)}. \tag{2.23}$$

This connects the observed residual phase modulation with established zeta fluctuations rather than with an unexplained new random source. The finite block coherence observed below is interpreted as the accumulation of these Selberg-scale phase defects under the slowly varying logarithmic clock.

2.6 Mathematical-physics interpretation and spectral systems

This section records the physical link. The article does not assert that the zeta zeros are the measured spectrum of a known laboratory Hamiltonian. The claim is that the construction defines a spectral observable naturally expressed in the same language used for quantum chaotic spectra.

(i) **Quantum chaotic spectra.** The identification $E_k = \gamma_k$ places the zeta ordinates in the same formal role as energy levels. The BGS conjecture relates chaotic quantum spectra to random-matrix statistics [9]; the zeta zeros enter this circle through the Montgomery–Odlyzko/Dyson connection [5, 6, 14].

(ii) **Trace-like spectral amplitudes.** The sum S_N^{ζ} is analogous to a finite spectral phase amplitude, except that the time is τ_k rather than a common time t. This is why the observable is closer to ordered spectral interference than to a spacing histogram.

(iii) **Semiclassical background plus spectral defect.** The RVM sequence is treated as a smooth Weyl-type background. The residual d_k is the phase response to the fluctuation δE_k, as shown by the exact residual response identity and the linear-response heuristic.

(iv) **Effective scale geometry.** Section 3.7 rewrites τ_k as a redshift factor in the scale coordinate $u = \log k$. This gives a formal link with phase-walk and redshift language, without assuming a physical spacetime metric.

2.7 Data source and computational protocol

The numerical study uses the first 2,001,052 zeta-zero ordinates from Odlyzko's public zero tables, distributed in the commonly used `zeros6` data set and documented as accurate to about $4 \cdot 10^{-9}$ in the ordinate values [15, 16]. The smooth comparison sequence is the corresponding Lambert-based RVM inverse-counting background constructed from the same index range. No private filename is part of the mathematical definition. A reproducible implementation needs only the public zero ordinates, the background construction, the logarithmic clock τ_k, and the figure-generation code.

3 Results and discussion

3.1 Finite cumulative geometry

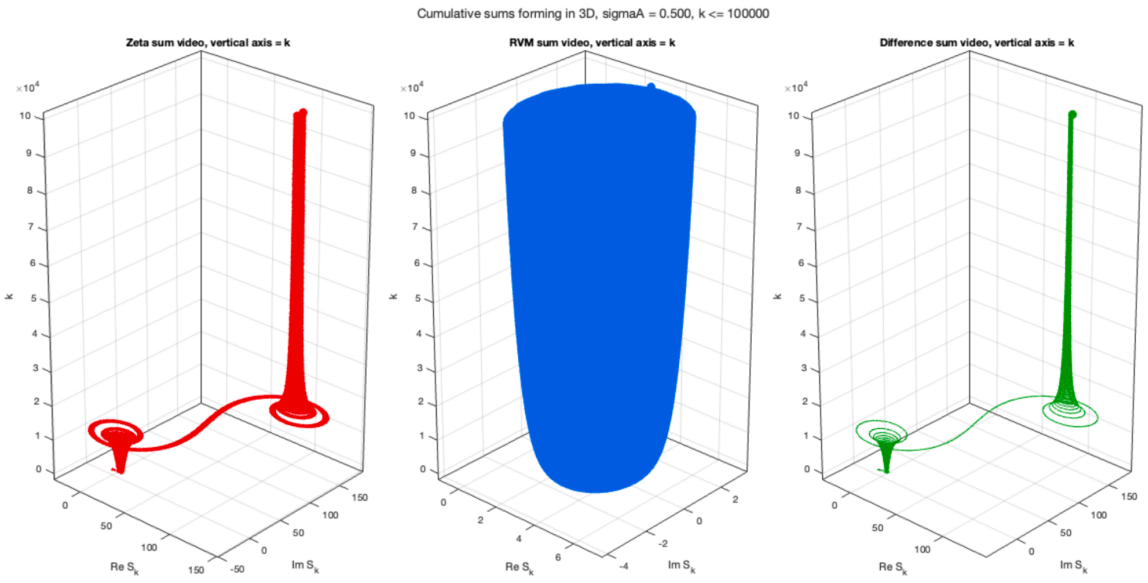


Figure 1: Ordered cumulative sums for $k \leq 100000$. The zeta curve is the primary object; the RVM curve is a smooth control; the difference curve isolates the residual contribution.

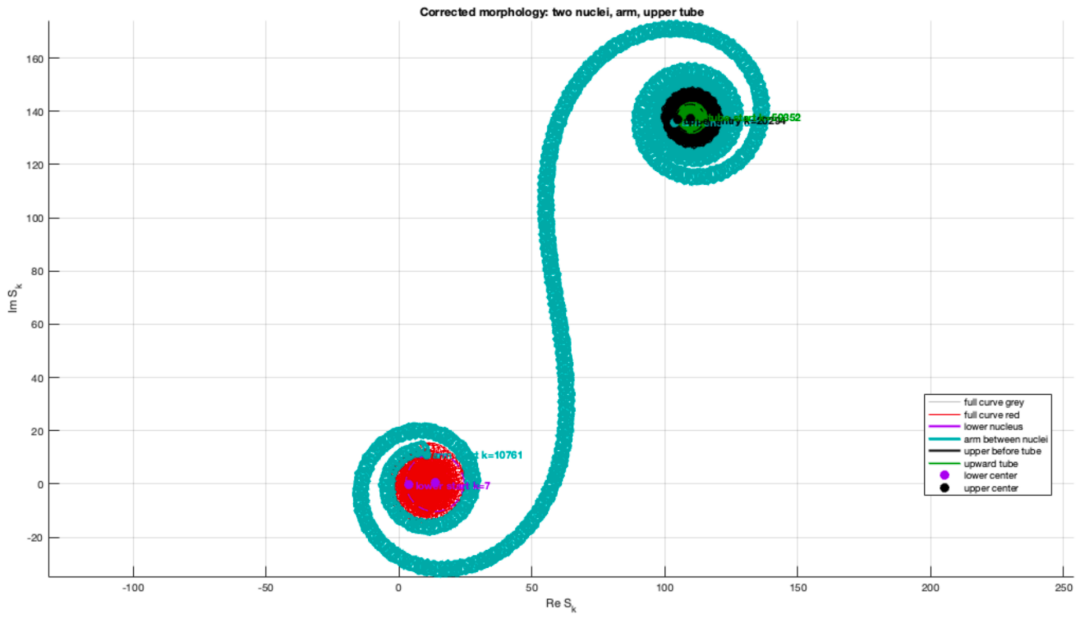


Figure 2: Two-dimensional morphology of the zeta cumulative curve. The labels are descriptive finite-regime names: lower nucleus, arm, upper entry, and upper tube.

Definition 3.1 *For the finite curve S_N^ζ, we use the descriptive names lower nucleus, arm, upper entry, and upper tube for the visible regions shown in Figure 2.*

The names in Definition 3.1 are finite descriptive labels. They are not asymptotic theorems.

Experiment 3.2 *For $N = 2000000$, the finite computation gives*

$$S_N^\zeta = 108.3327 + 133.6213i, \tag{3.1}$$

$$S_N^{\mathrm{RVM}} = 1.1510 - 4.4564i, \tag{3.2}$$

$$S_N^{\mathrm{diff}} = 107.1817 + 138.0777i. \tag{3.3}$$

The corresponding moduli are

$$|S_N^\zeta| = 172.0192, \tag{3.4}$$

$$|S_N^{\mathrm{RVM}}| = 4.6027, \tag{3.5}$$

$$|S_N^{\mathrm{diff}}| = 174.7952. \tag{3.6}$$

The final modulus ratios are

$$\frac{|S_N^{\mathrm{RVM}}|}{|S_N^\zeta|} = 0.02676, \tag{3.7}$$

$$\frac{|S_N^{\mathrm{diff}}|}{|S_N^\zeta|} = 1.01614. \tag{3.8}$$

3.2 Ordered window amplitudes

Definition 3.3 *For $x \in \{\zeta, \mathrm{RVM}, \mathrm{diff}\}$, define the window amplitude*

$$B_W^x(k) = \sum_{j=k}^{k+W-1} z_j^x. \tag{3.9}$$

For $x = \mathrm{diff}$, this means

$$B_W^{\mathrm{diff}}(k) = \sum_{j=k}^{k+W-1} d_j. \tag{3.10}$$

Definition 3.4 *The random-walk normalized window excess is defined by*

$$R_W^x(k) = \frac{|B_W^x(k)|}{\sqrt{W}\,\mathrm{rms}_{j\in[k,k+W-1]}(z_j^x)}. \tag{3.11}$$

Definition 3.5 *The window cancellation statistic is defined by*

$$C_W^x(k) = 1 - \frac{|B_W^x(k)|}{\sum_{j=k}^{k+W-1} |z_j^x|}. \tag{3.12}$$

Values $R_W^x(k)$ of order one are compatible with a random-walk scale. Larger values indicate stronger coherent alignment inside the ordered block. Values $C_W^x(k)$ close to one indicate strong cancellation, while smaller values indicate stronger internal alignment.

Experiment 3.6 *For the first $N = 2000000$ ordinates used in the computation, the residual window excess gives*

$$\max_k R_{500}^{\mathrm{diff}}(k) = 8.4750, \tag{3.13}$$

$$\max_k R_{1000}^{\mathrm{diff}}(k) = 11.8480, \tag{3.14}$$

$$\max_k R_{2000}^{\mathrm{diff}}(k) = 15.8891, \tag{3.15}$$

$$\max_k R_{5000}^{\mathrm{diff}}(k) = 11.0871. \tag{3.16}$$

The corresponding windows lie in the finite transition region near $k = 1.3 \cdot 10^4$ to $1.5 \cdot 10^4$.

Experiment 3.7 *For comparison, the RVM background gives*

$$\max_k R_{500}^{\mathrm{RVM}}(k) = 0.3040, \tag{3.17}$$

$$\max_k R_{1000}^{\mathrm{RVM}}(k) = 0.2172, \tag{3.18}$$

$$\max_k R_{2000}^{\mathrm{RVM}}(k) = 0.1561, \tag{3.19}$$

$$\max_k R_{5000}^{\mathrm{RVM}}(k) = 0.09884. \tag{3.20}$$

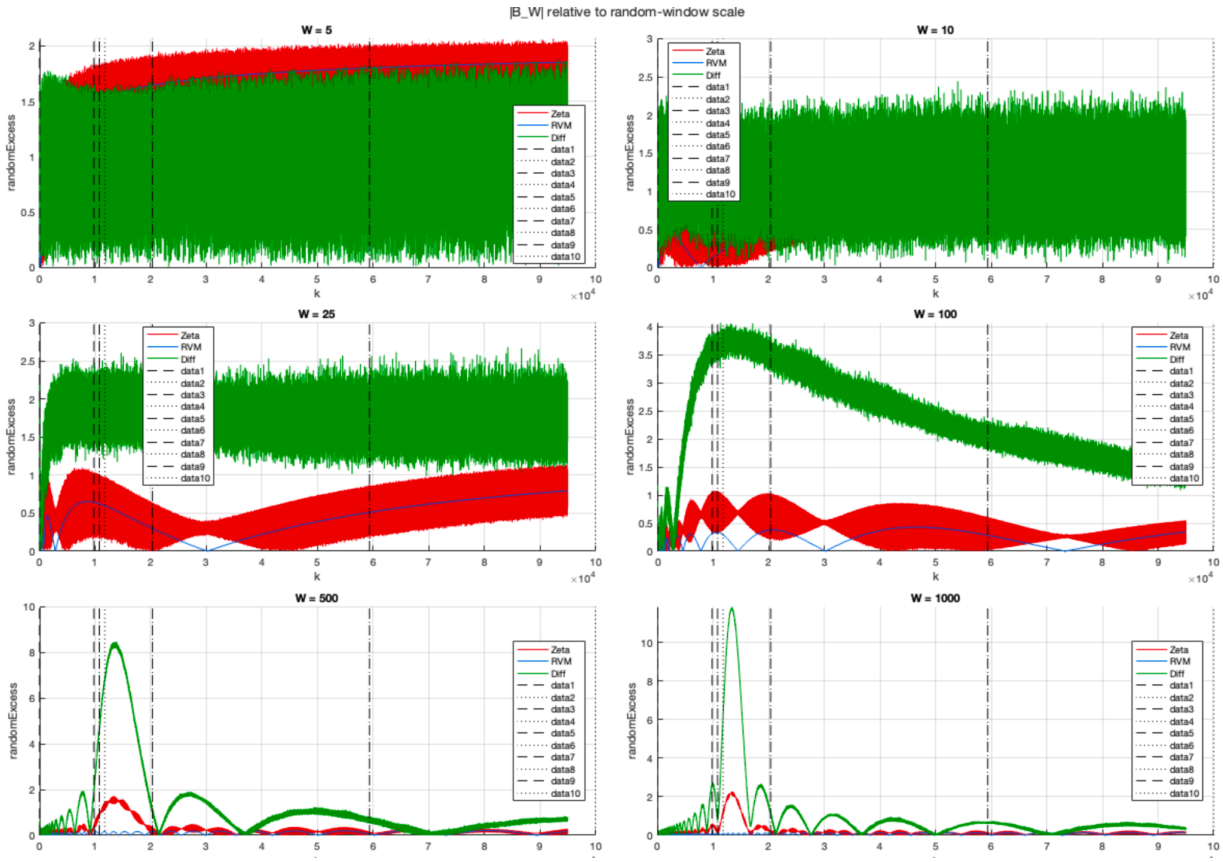


Figure 3: Local window diagnostics for the logarithmic phase-walk transform. The panels compare zeta, RVM, and residual window behaviour. The key point is that RVM is not flat: it shows the smooth susceptibility skeleton imposed by τ_k, while the zeta and residual signals excite that skeleton more strongly.

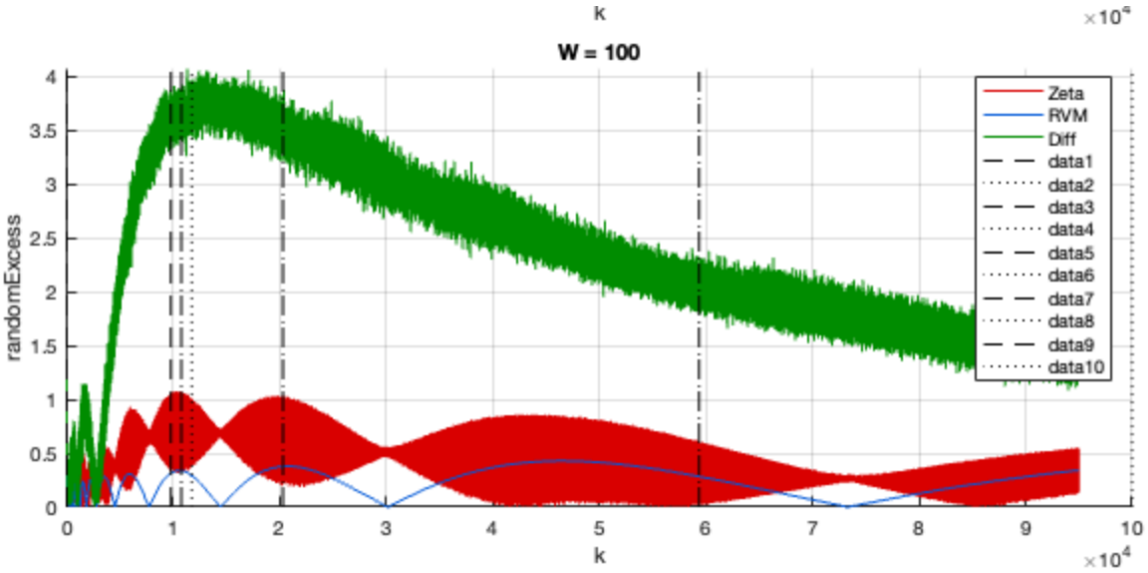


Figure 4: The $W = 100$ window panel. This figure resolves the early finite region where the arm begins to form. The residual and zeta curves show stronger excitation than the RVM control.

Heuristic 3.8 *This is a heuristic argument. For any finite sequence z_j, one has*

$$|B_W(k)|^2 = \sum_{a=0}^{W-1} \sum_{b=0}^{W-1} z_{k+a}\overline{z_{k+b}}. \tag{3.21}$$

Separating diagonal and off-diagonal terms gives

$$|B_W(k)|^2 = \sum_{a=0}^{W-1} |z_{k+a}|^2 + 2\,\mathrm{Re} \sum_{0\le a<b\le W-1} z_{k+a}\overline{z_{k+b}}. \tag{3.22}$$

The diagonal part sets the random-walk scale. The off-diagonal part determines whether the ordered block is coherently enhanced or suppressed. In the present transform, the phase differences are

$$\theta^{\zeta}_{k+a} - \theta^{\zeta}_{k+b} = \gamma_{k+a}\tau_{k+a} - \gamma_{k+b}\tau_{k+b}. \tag{3.23}$$

Thus the window excess probes ordered phase correlations of the logarithmic phase-walk field.

3.3 Robustness: Null test with synthetic Selberg-scale noise

Question 3.9 *Does independent Gaussian noise of the same scale as the measured spectral defect reproduce the controlled zeta morphology, or does it produce a substantially larger response under the logarithmic transform?*

The answer is tested by a null model: inject independent Gaussian noise with the same mean and standard deviation as the measured spectral defect $\delta E_k = \gamma_k - \gamma_k^{\mathrm{RVM}}$ into the smooth RVM sequence, apply the same logarithmic phase-walk transform, and compare both window amplitudes and final cumulative amplitudes.

Definition 3.10 *For the RVM ordinates γ_k^{RVM}, construct a synthetic perturbed sequence*

$$\gamma_k^{noisy} = \gamma_k^{RVM} + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2), \tag{3.24}$$

where $\mu_\delta = \mathrm{mean}(\gamma_k - \gamma_k^{RVM})$ and $\sigma_\delta^2 = var(\gamma_k - \gamma_k^{RVM})$ are the empirical mean and variance of the true spectral defect.

Experiment 3.11 *For the null-test phase-walk visualization, $N = 800$ is used only to keep the three displayed paths readable. This should not be confused with the main numerical data set, which contains 2,001,052 Odlyzko zero ordinates. On this displayed segment, the spectral defect statistics are*

$$\mu_\delta = -0.727324, \tag{3.25}$$
$$\sigma_\delta = 0.406515. \tag{3.26}$$

Apply the logarithmic phase-walk transform to three sequences: (i) γ_k (true zeta), (ii) γ_k^{RVM} (pure RVM), and (iii) γ_k^{noisy} (RVM + synthetic Gaussian noise of scale $(\mu_\delta, \sigma_\delta)$). Compute final cumulative amplitudes and window amplitudes:

$$|S_N^\zeta| = 13.004641, \tag{3.27}$$
$$|S_N^{RVM}| = 5.263621, \tag{3.28}$$
$$|S_N^{noisy}| = 27.426514. \tag{3.29}$$

For window amplitude $\max_k |B_W^x(k)|$ with $W = 500$:

$$RVM\ pure : 3.50, \tag{3.30}$$
$$RVM + noise : 27.78, \tag{3.31}$$
$$Zeta\ real : 11.67. \tag{3.32}$$

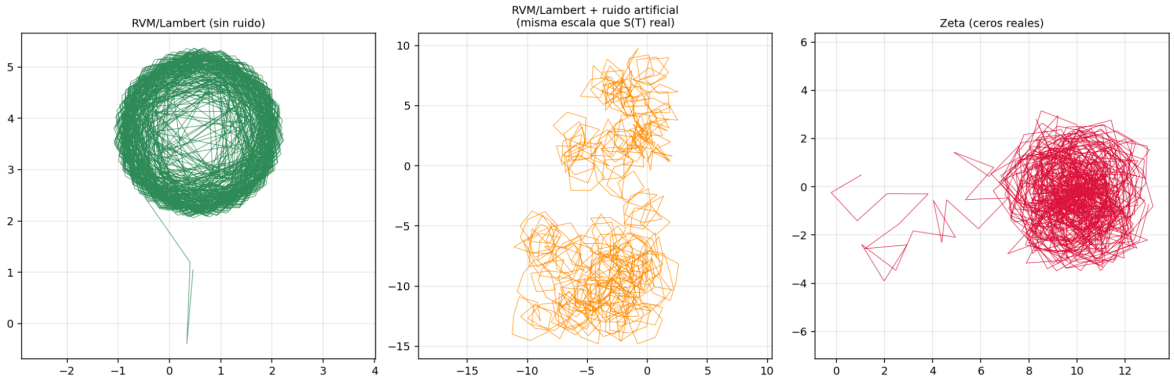


Figure 5: Null test for independent phase noise. Left: the pure RVM background is confined and smooth. Center: adding independent Gaussian noise of the same scale as the measured zeta–RVM spectral defect produces a substantially larger logarithmic phase-walk response. Right: the true zeta zeros produce an irregular but controlled response. For $W = 500$, the maximum window amplitudes are 3.50 for pure RVM, 27.78 for RVM plus independent noise, and 11.67 for the true zeta sequence. The final cumulative moduli are 5.26, 27.43, and 13.00, respectively. The null model therefore shows that unstructured phase noise can trigger the transform but gives a larger response than the zeta sequence; the observed zeta spirals and finite windows are controlled by the ordered zeta phase defect rather than by independent noise alone.

The null test should be read in the opposite way from a naive noise explanation. Independent Gaussian perturbations of the correct one-step scale do not simply reproduce the zeta response. They produce a substantially larger response under the logarithmic transform than the true zeta sequence. In the finite test reported above,

$$\max_k |B_{500}^{\mathrm{RVM}}(k)| = 3.50, \tag{3.33}$$

$$\max_k |B_{500}^{\mathrm{noise}}(k)| = 27.78, \tag{3.34}$$

$$\max_k |B_{500}^{\zeta}(k)| = 11.67, \tag{3.35}$$

and

$$|S_N^{\mathrm{RVM}}| = 5.26, \tag{3.36}$$

$$|S_N^{\mathrm{noise}}| = 27.43, \tag{3.37}$$

$$|S_N^{\zeta}| = 13.00. \tag{3.38}$$

Thus the logarithmic transform is sensitive to phase noise of the correct scale, but the zeta sequence is not behaving as independent Gaussian noise. The true zeta response is intermediate and controlled: much larger than the smooth RVM background, but substantially smaller than the independently perturbed RVM sequence. This supports the interpretation that the visible spirals and finite window structures are produced by zeta-specific ordered correlations in the phase defect, not by unstructured noise alone.

Heuristic 3.12 *This is a heuristic interpretation. The null model shows two separate effects. First, fluctuations of the correct scale are sufficient to activate the logarithmic clock. Second, if those fluctuations are made independent, the response is too large. The true zeta sequence therefore appears to combine Selberg-scale phase fluctuations with correlations that regulate the accumulated amplitude. In this sense the null test does not reduce the observed spirals to noise; it separates mere scale sensitivity from the ordered zeta correlations that control the final morphology.*

3.4 Phase portraits and local drift

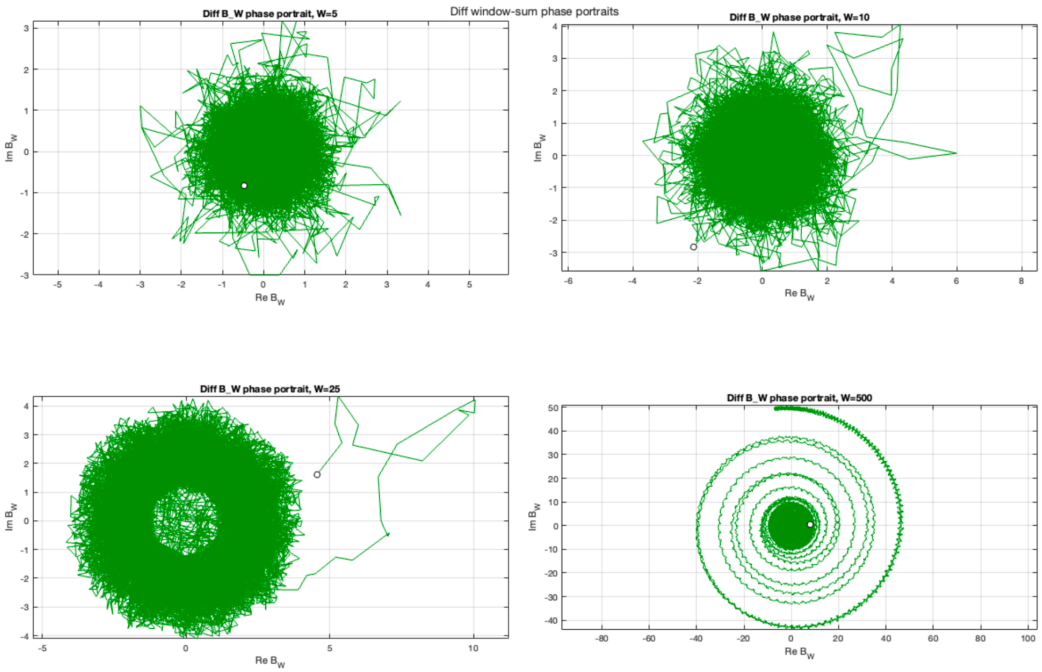


Figure 6: Phase portraits and local drift for selected window sums. The residual windows form organized loops rather than featureless scatter, and the local drift concentrates near the same finite transition region identified by the window diagnostics.

The phase portraits are not independent assumptions. They are local projections of the same ordered window amplitudes defined in (3.9). Their role is to show that the residual sequence is not featureless after ordered summation.

3.5 Nearest-neighbor gaps of the transformed function

Definition 3.13 *The direct vector gaps of the transformed zeta and RVM terms are*

$$D_k^{\zeta} = |z_{k+1}^{\zeta} - z_k^{\zeta}|, \tag{3.39}$$

$$D_k^{\mathrm{RVM}} = |z_{k+1}^{\mathrm{RVM}} - z_k^{\mathrm{RVM}}|. \tag{3.40}$$

The residual vector gap is

$$D_k^{\mathrm{diff}} = |d_{k+1} - d_k|. \tag{3.41}$$

Definition 3.14 *The angular functional gaps are*

$$\Delta\theta_k^{\zeta} = \gamma_{k+1}\tau_{k+1} - \gamma_k\tau_k, \tag{3.42}$$

$$\Delta\theta_k^{\mathrm{RVM}} = \gamma_{k+1}^{\mathrm{RVM}}\tau_{k+1} - \gamma_k^{\mathrm{RVM}}\tau_k, \tag{3.43}$$

$$\Delta\theta_k^{\mathrm{resid}} = \Delta\theta_k^{\zeta} - \Delta\theta_k^{\mathrm{RVM}}. \tag{3.44}$$

Definition 3.15 *For a local unfolding window* $W_{\mathrm{norm}} = 2000$*, define*

$$s_k^{\mathrm{func},\zeta} = \frac{\Delta\theta_k^{\zeta}}{\mathrm{movmedian}(\Delta\theta^{\zeta}, W_{\mathrm{norm}})}, \tag{3.45}$$

$$s_k^{\mathrm{func,RVM}} = \frac{\Delta\theta_k^{\mathrm{RVM}}}{\mathrm{movmedian}(\Delta\theta^{\mathrm{RVM}}, W_{\mathrm{norm}})}. \tag{3.46}$$

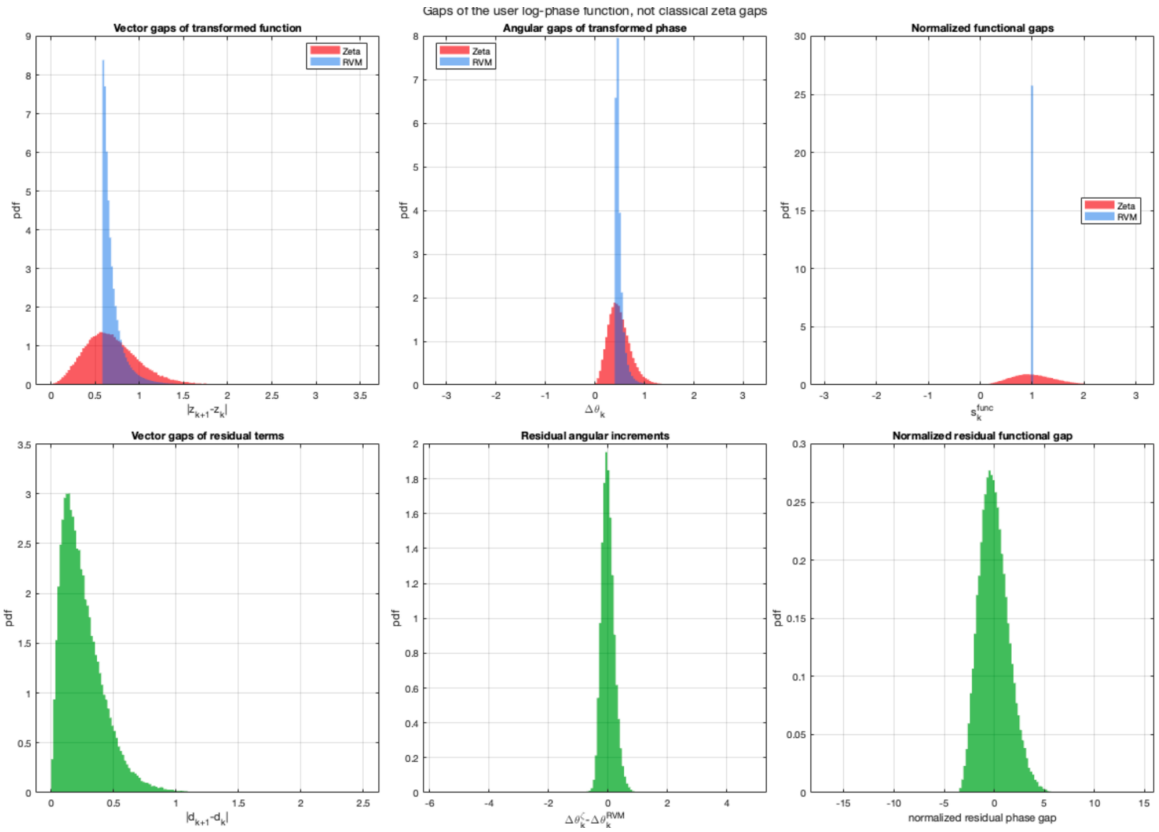


Figure 7: Nearest-neighbor gap diagnostics for the transformed function. The zeta functional gaps are much broader than the RVM control, while the residual one-step vector gap has its own radial law.

Experiment 3.16 *For* $N = 100000$ *and* $W_{\mathrm{norm}} = 2000$*, the direct vector-gap statistics include*

$$\mathrm{mean}(D^{\zeta}) = 0.6997, \tag{3.47}$$

$$\mathrm{mean}(D^{\mathrm{RVM}}) = 0.7048, \tag{3.48}$$

$$\mathrm{mean}(D^{\mathrm{diff}}) = 0.2569. \tag{3.49}$$

The standard deviations are

$$\mathrm{std}(D^{\zeta}) = 0.3374, \tag{3.50}$$

$$\mathrm{std}(D^{\mathrm{RVM}}) = 0.1720, \tag{3.51}$$

$$\mathrm{std}(D^{\mathrm{diff}}) = 0.1754. \tag{3.52}$$

For the locally normalized functional angular gaps,

$$\mathrm{std}(s^{\mathrm{func},\zeta}) = 0.4470, \tag{3.53}$$

$$\mathrm{std}(s^{\mathrm{func,RVM}}) = 0.04462. \tag{3.54}$$

Thus the zeta functional gaps have approximately ten times the standard deviation of the RVM control after local normalization.

3.6 Rayleigh local residual gaps

Definition 3.17 *The mean-normalized residual vector gap is*

$$s_k^{\mathrm{diffvec}} = \frac{|d_{k+1} - d_k|}{\mathrm{mean}_j |d_{j+1} - d_j|}. \tag{3.55}$$

Definition 3.18 *The comparison laws used in the nearest-neighbor residual test are*

$$p_{\mathrm{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}, \tag{3.56}$$

$$p_{\mathrm{Poisson}}(s) = e^{-s}, \tag{3.57}$$

$$p_{\mathrm{Rayleigh}}(s) = \frac{s}{\sigma_R^2} \exp\left(-\frac{s^2}{2\sigma_R^2}\right), \qquad \sigma_R = \sqrt{\frac{2}{\pi}}. \tag{3.58}$$

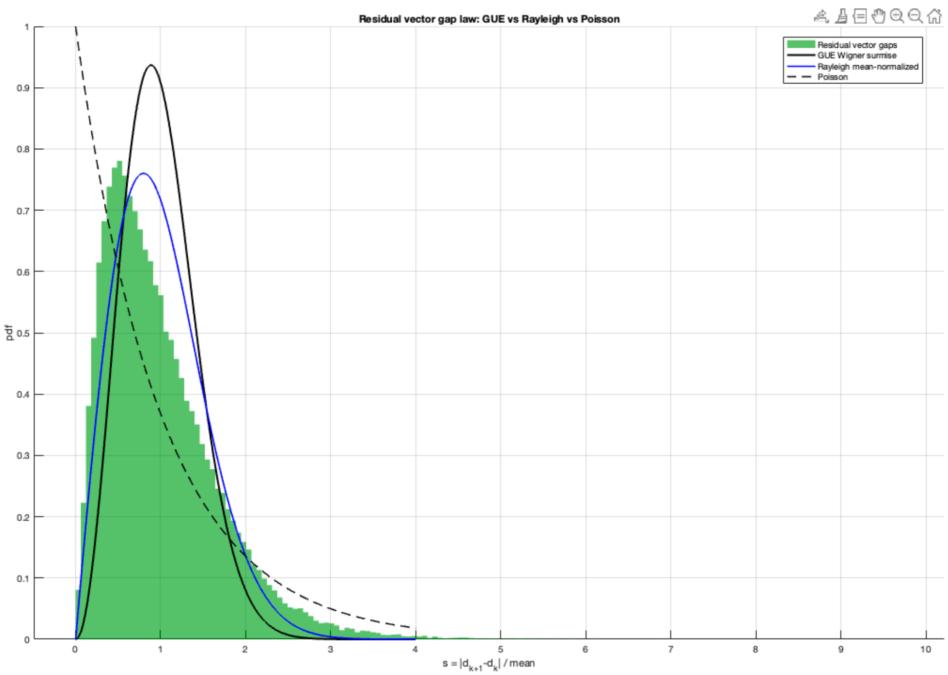


Figure 8: Nearest-neighbor residual vector gaps of the transformed function. After normalization by their mean, the residual gaps $|d_{k+1} - d_k|$ are compared with the GUE Wigner surmise, Poisson spacing, and the mean-normalized Rayleigh law.

Experiment 3.19 *For $N = 100000$, the mean raw residual vector gap is*

$$\mathrm{mean}_k |d_{k+1} - d_k| = 0.2569, \tag{3.59}$$

and the median normalized residual vector gap is

$$\mathrm{median}(s^{\mathrm{diffvec}}) = 0.8493. \tag{3.60}$$

The Kolmogorov–Smirnov distances are

$$\mathrm{KS}_{\mathrm{GUE}} = 0.2752, \tag{3.61}$$
$$\mathrm{KS}_{\mathrm{Rayleigh}} = 0.08390, \tag{3.62}$$
$$\mathrm{KS}_{\mathrm{Poisson}} = 0.1555. \tag{3.63}$$

The near-zero CDF exponent, fitted from

$$P(s \le \varepsilon) \sim \varepsilon^{\alpha}, \tag{3.64}$$

is

$$\alpha = 1.963. \tag{3.65}$$

The value (3.65) is close to the Rayleigh CDF exponent 2, not to the GUE nearest-neighbor exponent 3, and not to the Poisson exponent 1. Thus the one-step residual vector gaps are Rayleigh-like, not GUE-like.

Heuristic 3.20 *The Rayleigh law and the GUE spacing law describe different statistics. GUE spacing tests level repulsion and pair correlations in the zero ordinates themselves. The quantity $|d_{k+1} - d_k|$ is instead a one-step modulus of a complex residual phase difference after the logarithmic phase-walk transform. If the local real and imaginary components of this transformed residual are approximately centered Gaussian components, as one expects from a Selberg-scale phase fluctuation passed through a smooth complex background, then its modulus is Rayleigh-like. Thus Rayleigh behaviour here does not contradict the Montgomery–Odlyzko GUE picture. It says that the one-step residual is locally planar, while the window statistics test ordered block accumulation.*

Heuristic 3.21 *The residual difference $d_{k+1} - d_k$ is a complex planar quantity. If its real and imaginary components behave approximately as centered Gaussian components at the one-step level, then its modulus is expected to follow a Rayleigh-type radial law. Experiment 3.19 is consistent with this interpretation.*

3.7 Optional scale-clock interpretation

Heuristic 3.22 *This is only a heuristic reparametrization of the clock. If one introduces*

$$u = \log k, \tag{3.66}$$

then the logarithmic proper time becomes

$$\tau_k = \pi \frac{\log u}{u}. \tag{3.67}$$

One may formally call this a scale-dependent clock or an effective redshift factor. No result in this article is derived from gravitational field equations, and no physical spacetime metric is proposed.

The scale-clock language is optional. Its only role is to make clear that the observable evolves different spectral levels with different effective times. The numerical claims of the article are the definitions of the transform, the comparison with the RVM control, and the finite window and gap diagnostics.

3.8 Finite computational observation and open tests

Experiment 3.23 *For the first $N = 2000000$ ordinates in the finite computation, the residual window excess values in Experiment 3.6 imply the finite observations*

$$\max_k R^{\mathrm{diff}}_{500}(k) = 8.4750, \tag{3.68}$$

$$\max_k R^{\mathrm{diff}}_{1000}(k) = 11.8480, \tag{3.69}$$

$$\max_k R^{\mathrm{diff}}_{2000}(k) = 15.8891. \tag{3.70}$$

These are finite output values, not a theorem about all heights.

Question 3.24 *What is the distribution of* $\max_k R_W^{\mathrm{diff}}(k)$ *under null models that preserve the local size of* δE_k *but destroy the ordered phase structure? This question is necessary for deciding whether the observed window maxima represent more than the expected extreme values of a scanned correlated process.*

Question 3.25 *The computation up to* $N = 2000000$ *shows a stabilized upper-tube regime after the early transitions. Do further phase-block reorganizations occur at larger heights, producing new arms, tubes, or relocations of the cumulative geometry?*

3.9 Auxiliary real-coordinate response profile

The logarithmic phase-walk observable studied in the preceding sections uses the ordinate coordinate γ_k. This section records a separate construction involving the real coordinate σ. Its role is different from the phase-walk observable: it is a signed real-coordinate response built around the critical coordinate $1/2$. It is defined without assuming RH. If an actual zero is supplied with real part $\sigma_k \neq 1/2$, the associated signed weight records that displacement from the critical line. If RH holds, the supplied real-coordinate displacement is zero for every nontrivial zero and the signed weight vanishes term by term in the natural limiting sense. The parameter β in $D_X(\beta)$ is a supplied real coordinate. Therefore a plot of $D_X(\beta)$ as β varies should be read as a response profile of the construct, not as a statement that zeros exist at those externally chosen real parts. This is the correct scope of the object. The construct excludes, by definition, any off-critical contribution unless an off-critical real coordinate is supplied. Its potential value is as a formal starting point for analytic refinements involving the real coordinate, not as a standalone numerical search procedure.

3.10 Completed zeta function and local normalization

Definition 3.26 *The completed zeta function is*

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{3.71}$$

and satisfies the functional equation

$$\xi(s) = \xi(1-s). \tag{3.72}$$

For real σ and γ, the functional equation and the reality of ξ on the real axis give

$$|\xi(\sigma + i\gamma)| = |\xi(1 - \sigma + i\gamma)|. \tag{3.73}$$

This is a standard symmetry of the completed zeta function, not a detector by itself.

Definition 3.27 *Write a nontrivial zero as*

$$\rho_k = \sigma_k + i\gamma_k, \tag{3.74}$$

ordered by nondecreasing positive ordinate and counted with multiplicity. Its real displacement from the critical line is

$$\delta_k = \sigma_k - \frac{1}{2}. \tag{3.75}$$

Definition 3.28 *Let m_k be the multiplicity of the zero ρ_k of ξ, and let*

$$a_k = \frac{\xi^{(m_k)}(\rho_k)}{m_k!}. \tag{3.76}$$

Thus $a_k \neq 0$, and locally

$$\xi(s) = a_k(s-\rho_k)^{m_k} + O\left((s-\rho_k)^{m_k+1}\right). \tag{3.77}$$

For a supplied real displacement $\delta \neq 0$, define the multiplicity-normalized coefficient

$$C_k(\delta) = \left|\frac{\xi(\rho_k+\delta)}{a_k\delta^{m_k}}\right|^{-2}, \tag{3.78}$$

whenever the expression is finite and nonzero. This coefficient is positive by construction whenever it is defined. In the simple-zero case $m_k = 1$, $a_k = \xi'(\rho_k)$, so (3.78) reduces to the usual first-derivative normalization.

3.11 Counting coordinate and response profile

Definition 3.29 *Let*

$$N(T) = \#\{k : 0 < \gamma_k \leq T\} \tag{3.79}$$

be the zero-counting function, with zeros counted according to the ordering convention above. At the ordinate γ_k, the finite counting coordinate is

$$u_k = \log N(\gamma_k) = \log k. \tag{3.80}$$

Definition 3.30 *For $X > 0$ and a supplied real coordinate β, set*

$$\delta = \beta - \frac{1}{2} \tag{3.81}$$

and define the response profile

$$D_X(\beta) = \sum_{\gamma_k \leq X} C_k(\delta)\left(N(\gamma_k)^{2\delta} - N(\gamma_k)^{-2\delta}\right). \tag{3.82}$$

Equivalently, using $N(\gamma_k) = k$,

$$D_X(\beta) = \sum_{k \leq N(X)} C_k(\delta)\left(k^{2\delta} - k^{-2\delta}\right). \tag{3.83}$$

Since $u_k = \log k$, each signed weight difference is

$$k^{2\delta} - k^{-2\delta} = 2\sinh(2\delta u_k). \tag{3.84}$$

Consequently, for fixed externally supplied δ, the sign of each defined nonzero summand is the sign of δ. This sign property is algebraic and contains no information about whether a zero actually has real part β.

If β is chosen externally, for example $\beta = 0.9$, the sign of the summands is dictated by the supplied displacement $\beta - 1/2$. Such a calculation describes the response of the formal real-coordinate profile at that supplied coordinate. It is different from evaluating the construct on an actual zero with known real part σ_k. The distinction is essential: an externally supplied β probes the profile; an actual σ_k records the real-coordinate displacement of a supplied zero.

3.12 Critical involution and the role of the actual real parts

Under the critical involution $s \mapsto 1 - s$, the real displacement changes sign,

$$\delta \longmapsto -\delta, \tag{3.85}$$

while the counting coordinate remains the positive coordinate

$$u_k = \log k > 0. \tag{3.86}$$

Thus the involution exchanges

$$e^{2\delta u_k} \longleftrightarrow e^{-2\delta u_k} \tag{3.87}$$

by changing δ, not by replacing the actual coordinate u_k with $-u_k$.
It is not correct to identify the actual counting coordinate with a transformation $u_k \mapsto -u_k$. The equality $u_k = -u_k$ would force $u_k = 0$, which is not the case for $k > 1$. The only valid exchange is the weight exchange in (3.87) at fixed positive u_k.

Definition 3.31 *If a zero $\rho_k = \sigma_k + i\gamma_k$ is supplied with its actual real coordinate, define its actual real displacement by* (3.75)*. The associated per-zero signed response is defined by*

$$\mathcal{D}_k = 0 \qquad \text{if } \delta_k = 0, \tag{3.88}$$

and by

$$\mathcal{D}_k = C_k(\delta_k)\left(k^{2\delta_k} - k^{-2\delta_k}\right) \qquad \text{if } \delta_k \neq 0, \tag{3.89}$$

whenever the coefficient is defined. This quantity uses the actual σ_k, not an externally chosen β.

If RH holds, then $\delta_k = 0$ for every nontrivial zero and $\mathcal{D}_k = 0$ term by term by (3.88). If a zero is supplied with $\sigma_k \neq 1/2$, then $\delta_k \neq 0$ and (3.89) gives a signed response whose sign records the side of the critical line on which that supplied zero lies. Thus the construct is built without assuming RH, and it would respond immediately to an off-critical real coordinate if such a zero were supplied.
The article therefore contains two complementary coordinates. The logarithmic phase-walk observable uses γ_k and reveals ordered phase geometry in the ordinate direction. The real-coordinate response profile uses σ_k as a supplied coordinate and records displacement from the critical line. In its present form this is a bookkeeping construct, not a complete analytic theory; its relevance is that it gives a clean starting point for possible refinements in which the real-coordinate response is connected to deeper analytic structure of ξ.

4 Conclusions

We have defined and analyzed a logarithmic phase-walk observable for zeta-zero ordinates, compared it with the RVM background, and demonstrated its structural properties on the first 2,001,052 Odlyzko zero ordinates. The null-test visualization is included only as a qualitative illustration of the contrast between the smooth RVM control, independently perturbed RVM data, and the ordered zeta response; it is not the scale of the main computation.

Main findings: The one-step residual vector gaps follow a Rayleigh law ($\mathrm{KS}_{\mathrm{Rayleigh}} = 0.084$) rather than the GUE Wigner-surmise comparison ($\mathrm{KS}_{\mathrm{GUE}} = 0.275$). This Rayleigh behaviour is observed in a different object from the usual Montgomery–Odlyzko statistic: GUE describes the spacings of the zero ordinates, while the Rayleigh law appears in the radial distribution of one-step complex residual increments after the logarithmic phase-walk transform. This is the first reported detection of such Rayleigh-type behaviour for this transformed zeta–RVM residual observable. The coherent signal emerges at the level of ordered window amplitudes, with residual window excess reaching factors of 8–16 above the random-walk scale for $W = 500$–2000.

Mechanism. The observable's sensitivity to ordered phase fluctuations is rooted in the spectral defect $\delta E_k = \gamma_k - \gamma_k^{\mathrm{RVM}}$, which equals the classical oscillatory term $S(T)$ in the Riemann–von Mangoldt formula (as described in the null-test discussion). By Selberg's central limit theorem, $S(\gamma_k)$ has variance $\sim (1/2\pi^2)\log\log\gamma_k$, yielding δE_k of order $\sigma_\delta \approx 0.4$. The logarithmic phase-walk weight $\tau_k = \pi \log\log k/\log k$ then amplifies these known phase fluctuations through quasi-resonant summation. The null experiment shows a sharper distinction: independent perturbations of matched one-step scale produce a substantially larger response under the logarithmic transform, whereas the true zeta sequence gives a controlled intermediate response. Thus the logarithmic clock is sensitive to Selberg-scale fluctuations, but the observed spiral and window morphology require the ordered correlations present in the zeta phase defect.

Clarification on GUE. The GUE Wigner surmise describes level correlations between distant zeros of $\zeta(s)$—a global spectral property. The Rayleigh law observed here describes one-step differences of a residual function under nonlinear phase projection—a local planar property. These are complementary statistics of different objects and do not contradict each other. Both hold simultaneously.

Scope and generality. The construction is not limited to zeta. Any ordered quantum chaotic spectrum, random-matrix spectrum, or spectral model with a smooth background and a level-dependent clock can be analyzed by the same method. The observable reveals phase-interference structure that local spacing statistics discard because they ignore ordering.

Two-coordinate bookkeeping. Section 3.9 adds a complementary real-coordinate construct using the abscissa σ_k. It is defined independently of RH: on the critical line its signed response vanishes, while a supplied zero with $\sigma_k \neq 1/2$ gives a nonzero signed response. In its present form it is a modest formal device, not a complete analytic theory, but it supplies a clean point of departure for possible refinements involving the real coordinate of zeros. This complements the ordinate-based phase geometry studied in the rest of the paper.

Supplementary Material

The supplementary submission package contains the LaTeX source file and the compiled PDF manuscript. It also contains the separate PNG figure files cited in the manuscript.

Data Availability

The zero ordinates used in the main computation are the first 2,001,052 entries of Odlyzko's public `zeros6` table. The figures and numerical tables in this manuscript are generated from those public ordinates, the Lambert W-based RVM background described in the text, and the logarithmic phase-walk definitions stated above. The source TEX file and all figure files are included with the submission package.

Conflicts of Interest

The author declares that there are no conflicts of interest regarding the publication of this paper.

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